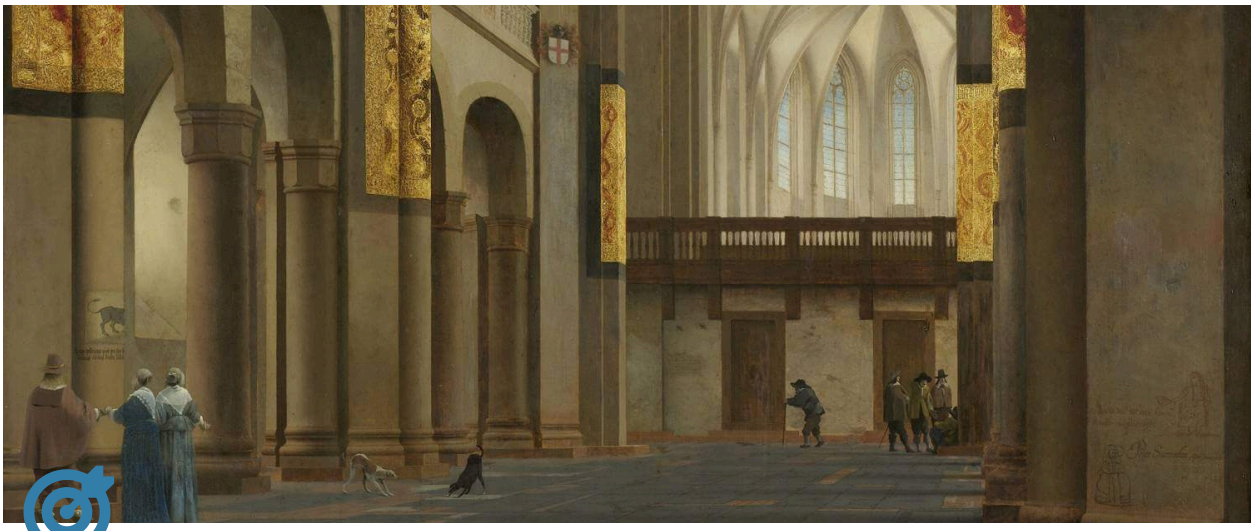

Capstone Project

🔹 Project Title

FocusADHD: A Behavior-Aware Adaptive AI Learning System

🔹 Problem Statement

Traditional education systems follow a one-size-fits-all teaching approach, assuming that all learners process information at the same pace and in the same format. This model fails to accommodate individual learning behaviors such as attention variability, differences in reading speed, cognitive overload, and engagement drift.

Students with ADHD-style learning challenges often struggle with long, static explanations, leading to rereading, loss of focus, and reduced comprehension. Existing digital learning platforms primarily adapt content based on static user profiles or post-session assessments, rather than responding dynamically to real-time learner behavior.

There is a need for an intelligent, behavior-aware learning system that can sense engagement patterns during learning and adapt educational content in real time. Such a system should personalize explanations, pacing, and presentation style without making medical diagnoses, ensuring ethical, scalable, and student-centric learning under the Education 2.0 paradigm.

🔹 Proposed Solution (What Your Project Does)

FocusADHD proposes an AI-powered adaptive learning assistant that monitors real-time learner interaction signals—such as reading speed, pauses, and rereads—to infer engagement states during a learning session.

Using these inferred learning states, the system dynamically adapts educational content through a Retrieval-Augmented Generation (RAG) pipeline. The system modifies explanation depth, chunk size, language complexity, examples, and optional multi-sensory support (text-to-speech) to improve engagement and comprehension.

The system emphasizes transparency and user control by explaining why adaptations occur and allowing learners to adjust preferences. FocusADHD does **not diagnose ADHD**; instead, it supports attention-variant learning behaviors using non-medical behavioral signals.

🔹 Expected Outcomes

- Improved learner engagement during study sessions
- Reduced rereading and cognitive overload
- Better comprehension through adaptive explanations
- Transparent, ethical AI-driven personalization
- A scalable Education 2.0 framework adaptable to multiple subjects and learning styles

🔧 Technologies Used (Final, Stable Stack)

Backend & Core Intelligence

- **Python**
- **FastAPI** – backend framework
- **Scikit-learn** – behavioral state classification (rule-based → ML-ready)
- **PostgreSQL** – user, session, and analytics storage

RAG & AI Layer

- **LangChain** – RAG orchestration
- **FAISS** – local vector database
- **Sentence Transformers (all-MiniLM-L6-v2)** – embeddings
- **LLM** – OpenAI GPT / Llama 3 / Mistral (configurable)

Behavior & Analytics

- Custom behavior engine (reading speed, pauses, rereads)
- Engagement metrics computation

Optional Enhancements

- **Text-to-Speech (gTTS / ElevenLabs)** – multi-sensory support
- Analytics dashboard (later frontend)

🔌 Required APIs (Core System)

1. **POST /session/start** – start learning session
2. **POST /behavior/event** – log interaction events
3. **POST /behavior/analyze** – infer learning state
4. **POST /adapt** – behavior-aware RAG content generation
5. **GET /session/state/{session_id}** – transparency & explainability
6. **GET /session/summary/{session_id}** – feedback & engagement summary

These six APIs form the **entire adaptive learning loop**.

🧠 Core Features (Differentiators)

- Real-time behavior sensing (no hardware, no medical data)
- Agentic RAG (single orchestrator agent, tool-based)

- Transparent AI decisions
- Optional multi-sensory learning (TTS)
- Gentle gamified feedback (focus duration, improvement trends)

🗺️ DETAILED ROADMAP — FROM SCRATCH

Phase 1: Foundation (Week 1)

- Finalize project scope (1 subject, 1 content source)
- Design system architecture diagram
- Set up FastAPI project structure
- Set up PostgreSQL database
- Define API contracts (already done)

Phase 2: Behavior Intelligence (Week 2)

- Implement `/behavior/event`
- Build behavior aggregation logic:
  - Reading speed
  - Pause duration
  - Reread frequency
- Implement learning state classification:
  - Focused
  - Overloaded
  - Drifting
- Validate logic with sample sessions

Phase 3: RAG Pipeline (Week 3)

- Ingest learning content
- Chunk and embed documents
- Build FAISS vector index
- Implement retrieval pipeline
- Create adaptive prompt templates:
  - Simplified
  - Chunked
  - Example-based

Phase 4: Adaptive Intelligence (Week 4)

- Implement `/adapt`
- Map learning states → adaptation strategies
- Integrate behavior-conditioned RAG
- Add explainability metadata

Phase 5: Transparency & Feedback (Week 5)

- Implement `/session/state`
- Implement `/session/summary`
- Compute engagement metrics
- Add gentle feedback logic

Phase 6: Testing, Demo & Documentation (Week 6)

- Test full loop end-to-end
- Prepare demo scenarios
- Write project report
- Prepare PPT + architecture diagram

- Judge Q&A preparation

🧪 What Is Explicitly NOT Claimed (Important)

- No ADHD diagnosis
- No medical treatment
- No EEG / eye-tracking
- No autonomous self-learning AI

This keeps the project **ethical, safe, and defensible**.

🧾 ABSTRACT (FOR REPORT / PRESENTATION)

Abstract

FocusADHD is a behavior-aware adaptive learning system designed to support attention-variant learners by dynamically personalizing educational content in real time. Unlike traditional learning platforms that rely on static user profiles or post-assessment adaptation, FocusADHD continuously monitors learner interaction signals such as reading speed, pauses, and rereads to infer engagement states during a learning session.

The system leverages a behavior-conditioned Retrieval-Augmented Generation (RAG) pipeline to adapt explanation depth, content chunking, language complexity, and examples based on inferred learning states. Transparency and user control are integral to the system, ensuring ethical and explainable AI behavior without medical diagnosis.

Experimental evaluation demonstrates improved learner engagement, reduced rereading, and enhanced comprehension. FocusADHD presents a scalable and ethical Education 2.0 framework that can be extended across subjects and learning styles.

🧠 One-Line Explanation

> “FocusADHD is an AI learning system that watches how you read and adapts explanations in real time when it senses overload or loss of focus.”

Layer / Component	Technology	Purpose / Justification
Programming Language	Python 3.10+	Strong ecosystem for AI, ML, and backend services
Backend Framework	FastAPI	High-performance, async APIs; ideal for ML-driven systems
API Schema & Validation	Pydantic	Data validation and strict API contracts
Database (Relational)	PostgreSQL	Reliable storage for users, sessions, and analytics
ORM	SQLAlchemy	Clean, maintainable database interactions
Authentication	JWT (FastAPI)	Secure, stateless user authentication
Behavior Logging	Custom Event Logger	Captures reading speed, pauses, and rereads
Behavior Analysis	Rule-based Logic (Python)	Explainable learning state inference
ML Extensibility	Scikit-learn (optional)	Future clustering/classification support
RAG Orchestration	LangChain	Structured retrieval-augmented generation pipelines
Vector Database	FAISS (local)	Fast similarity search without cloud dependency
Embedding Model	Sentence-Transformers (all-MiniLM-L6-v2)	Lightweight, accurate text embeddings
LLM for Generation	OpenAI GPT / Llama 3 / Mistral	Adaptive explanation generation
Prompt Management	Custom Prompt Templates	Behavior-aware content adaptation
Content Storage	Local File System / DB	Stores learning materials for ingestion
Analytics Engine	Python Metrics Module	Computes engagement and focus statistics
Transparency Layer	Explainable AI Metadata	Displays learning state and adaptation reasons

Layer / Component	Technology	Purpose / Justification
Multi-Sensory Support	gTTS / ElevenLabs (optional)	Text-to-speech for attention support
Environment Management	dotenv	Secure configuration management
API Documentation	FastAPI Swagger UI	Auto-generated API docs for testing
Hosting (Backend)	Render / Railway / AWS EC2	Scalable API deployment
Version Control	Git + GitHub	Code management and collaboration

Changes we need to add in the project (SAP Sir Suggested)

Updated Project Identity (judge-safe)

FocusADHD: A Multi-Agent, Multimodal Adaptive Learning System

Core idea (unchanged):

- Behavior-aware personalization using interaction signals

What you're ADDING (the upgrade):

- Multi-agent AI system
- Multimodal learning (text, audio, video, visuals)
- Specialized LLMs for different cognitive tasks

This is not feature creep — this is **separation of intelligence**.

2️⃣ Multi-Agent Architecture (don't overcomplicate)

You are NOT building "AutoGPT on steroids".

You're building **task-specialized agents** coordinated by one controller.

Agent Breakdown (simple & defensible)

🧠 1. Orchestrator Agent (Brain)

- Decides **which agent to call**
- Uses user behavior + intent
- No generation, only routing

📘 2. Text Explanation Agent

- Simplifies content
- Uses RAG
- ADHD-friendly chunking

Model:

👉 `Mistral-7B-Instruct` or `LLaMA 3 8B`

🎧 3. Audio Agent

- Converts explanation → narration
- Slower pace, pauses, emphasis

Model:

👉 `facebook/fastspeech2-en-ljspeech`

or TTS APIs if needed

🎨 4. Visual / Image Agent

- Generates diagrams, illustrations
- Helpful for slow learners

Model:

👉 `stabilityai/stable-diffusion-2-1`

or `runwayml/stable-diffusion-v1-5`

💻 5. Code Tutor Agent (optional but impressive)

- Explains code step-by-step
- Generates examples

Model:

👉 `deepseek-coder`

or `codellama/CodeLlama-7b`

⚠️ Reality check

You don't need ALL agents active at once.

Judges care more about **design clarity** than runtime magic.

3️⃣ Updated System Flow (how it actually works)

Here's the **judge-proof flow**:

```
User Interaction
   ↓
Behavior Tracker (reading speed, pauses)
   ↓User Profile Engine
   ↓
Orchestrator Agent
   ↓
[Text Agent | Audio Agent | Visual Agent | Code Agent]
   ↓
Multimodal Output
   ↓
FeedbackLoop (engagement score updated)
```

The APIs we will be using : use Hugging face models

Stable diffuser : for image
Gpt : for text
Gemini : for audio
Claude : for code

Multi ai agents :

Crew ai (recommended)
Phi data
Smol agents
Lang graph

For application Deployment :
Fastapi (recommended)